Insurance Claim Denial Investigation Framework

Structured investigation workflow for evaluating health insurance claim denials

Purpose. Organize claim-denial review into seven specialized categories, with evidence checks, policy-table lookups, and calculation steps.

Category 1: MEDICAL_NECESSITY

Core Question: Was this hospitalization or procedure clinically required, or could it have been avoided, treated conservatively, or performed as an outpatient?

1. Clinical Severity and Vital Instability Scan

- Scan the discharge summary and doctor's notes for objective indicators of physiological severity.
- Identify vital instability (for example, oxygen saturation below 90%), severe pain scores (≥ 8/10), and diagnostic severity grades (for example, Grade IV osteoarthritis with bone-on-bone contact or 90% LAD arterial stenosis).

2. Conservative Management Audit

- Search the patient's history for non-surgical and non-invasive therapies attempted before surgery: physiotherapy, steroid injections, pharmacotherapy, and lifestyle interventions.
- Duration = Date of Surgery − First Recorded Conservative Treatment Date.
- Check whether the duration meets the relevant clinical threshold, such as at least six continuous months.

3. Clinical Guideline Matching

- Cross-reference the ICD-10 diagnosis in icd10_codes.csv and the procedure in procedure_codes.csv against relevant medical-necessity criteria.
- Potential references include InterQual, Milliman Care Guidelines, and Indian clinical protocols.
- Check whether surgical indications were met, such as failed conservative therapy plus radiological evidence of end-stage joint destruction.

4. Physician Certification Verification

- Verify whether the treating surgeon issued a written certificate stating that inpatient surgery was medically necessary to prevent permanent disability, organ failure, or mortality.

Category 2: PRE_EXISTING_WAITING_PERIOD

Core Question: Did the disease originate before the policy began, and does the insurer have the legal right to repudiate the claim after considering continuous policy renewals?

1. First-Symptom and Onset Chronology Tracing

- Scan the History of Present Illness (HPI) and past consultation records for statements such as “symptoms for three years,” “hypertensive for five years,” or “first noticed swelling in 2021.”
- Extract the earliest documented onset date, D_onset.

2. Policy Timeline Calculation

- Elapsed policy time = Date of Hospital Admission − Policy Inception Date.
- Compare the elapsed time with the waiting periods configured in policy_clauses.csv.
- Example values in the source framework: initial waiting period—30 days; specified diseases (such as cataracts, hernia, and joint replacement)—24 months; pre-existing disease—36 or 48 months.

3. IRDAI Statutory Moratorium Clause Audit

- Check the number of continuous, uninterrupted policy-renewal years.
- The source framework describes an eight-year period and a five-year (60-month) period under IRDAI 2024 regulations. Verify the applicable current rule and its conditions before relying on this comparison.
- If the applicable moratorium threshold is met, flag a denial based on non-disclosure or pre-existing disease for legal review, subject to applicable exceptions and fraud provisions.

4. Proposal Form Non-Disclosure Cross-Check

- Compare the condition with the information declared on the original proposal form.
- Investigate whether the condition was asymptomatic and undiagnosed at policy inception. For example, hypertension discovered during a routine pre-operative checkup does not by itself establish pre-existing non-disclosure.

Category 3: POLICY_EXCLUSION

Core Question: Does the cited policy exclusion apply to the case, or is there an exception or sub-clause that overrides it?

1. Exact Clause Retrieval and Scope Verification

- Use vector search over policy_clauses.csv to retrieve the clause cited in the denial letter.
- Example clauses: Clause 4.3—Elective Surgeries; Clause 4.12—Genetic Disorders.
- Check whether the procedure code in procedure_codes.csv and ICD-10 code in icd10_codes.csv fall within the clause's scope.

2. Exception Clause Search

- Analyze the clause's has_exception and exception_text attributes.
- Look for conditional exceptions—for example, exclusions subject to accidental bodily injury, cancer-treatment necessity, or applicable provisions under Section 21 of the Mental Healthcare Act, 2017.

3. Exception Criteria Compliance Audit

- Evaluate the patient's clinical evidence against the exception criteria.
- If an applicable exception appears satisfied, flag a potential Misapplication of Exclusion for review.

Category 4: FINANCIAL_LIMIT

Core Question: Did the insurer calculate deductions correctly, or did it improperly apply deductions involving room rent, sub-limits, or non-medical items?

1. Room Rent Proportionate Deduction Audit

- Check whether the patient occupied a room category above the policy limit—for example, deluxe suite when eligible for twin-sharing accommodation.
- Deduction Ratio = Eligible Room Rent ÷ Actual Room Rent.
- Check whether proportionate deductions were applied to items that should not be scaled by room category, such as medicines or implants. Flag potentially improper deductions for review under applicable rules and policy terms.

2. Sub-Limit and Capping Verification

- Audit treatment-specific caps against the remaining Sum Insured (SI).
- Examples in the source framework: cataract treatment capped at ₹40,000 per eye, ICU charges capped at 2% of SI, and robotic-surgery caps.

3. Non-Medical Expenses (NME) Bill Audit

- Compare every item on the itemized hospital bill against nme_exclusions.csv, based on the configured IRDAI Annexure IV list.
- Examples of genuine non-medical items: admission kits and attendant food.
- Items to flag for review: surgical gloves, syringes, IV cannulas, pulse oximeter charges, and PPE kits during infectious-disease treatment.
- Potential reclaimed amount = Total Bill − Permissible NME − Approved Amount. Validate the formula against the claim's actual billing and approval definitions before use.

Category 5: DOCUMENTATION

Core Question: Are the documents requested by the insurer genuinely missing, or did the insurer overlook documents already submitted?

1. Deficiency Letter Parsing

- Parse the insurer's query letter or repudiation notice and extract the requested documents.
- Examples: Indoor Case Papers (ICP), Treating Doctor Certificate, Pre-Authorization Form, and original film X-rays.

2. Repository Gap Analysis

- Compare the insurer's requested list with the patient's uploaded document manifest.
- Already Available: The document exists in the archive; flag that the insurer may have overlooked submitted records.
- Genuinely Missing: The document was not uploaded or found in the available repository.

3. Hospital Action Item and Requisition Generator

- Identify the issuing entity: Hospital Records Department, Third-Party Administrator (TPA) desk, or treating physician.
- Draft a specific requisition letter for each genuinely missing record.

Category 6: ADMINISTRATIVE_PROCEDURAL

Core Question: Was the delay in claim intimation or hospital submission justifiable, and can the insurer legally repudiate the claim on procedural or technical grounds?

1. Filing and Intimation Deadline Calculation

- Intimation delay = Date of Intimation − Date of Hospital Admission.
- Submission delay = Date of Claim Submission − Date of Discharge.
- Compare the calculated durations against the policy clauses; example deadlines in the source framework are 24-hour intimation and 15-day submission.

2. IRDAI Statutory Delay Circular Evaluation

- Cross-reference the delay against IRDAI Circular Reference IRDA/HLT/MISC/CIR/216/09/2011.
- Evaluate whether the delay resulted from circumstances beyond the insured's control.
- Check records for emergency admission, ICU stay, loss of consciousness, or caregiver unavailability as possible grounds for explaining delay.

3. Non-Network Hospital and Emergency Protocol Check

- If the claim was denied because a non-network hospital was used without prior intimation, check whether admission involved an acute emergency.
- Verify applicable hospital requirements, including registered-bed criteria and round-the-clock doctor and nursing availability.

Category 7: ELIGIBILITY_COVERAGE

Core Question: Was the patient covered, and was the policy active on the date the medical service was provided?

1. Member Dependency and Age Eligibility Check

- Validate the insured member's relationship and age against the policy's eligibility criteria.
- Examples: dependent children covered up to age 25 or until marriage, where applicable, and dependent parents meeting entry-age requirements.

2. Premium Grace Period Audit

- If the claim was denied for policy lapse, compare the hospitalization date with the renewal due date and applicable grace period.
- Example test: Admission Date ≤ Renewal Due Date + 30 days.
- Verify whether the renewal premium was paid within the applicable grace period and whether continuity of coverage is preserved under the relevant regulations and policy terms.

3. Portability and Waiting Period Continuity Check

- For recently ported policies, check portability records to verify that eligible waiting-period credits from the previous insurer were carried forward.
- Flag cases where accrued credits may have been incorrectly removed.

Summary: What Makes This Architecture Powerful

The architecture performs category-specific investigations rather than relying on a generic AI prompt. Each denial type triggers a specialized evidence and rule review.

Denial Category	Specialized Investigation
Medical Necessity	Clinical severity assessment and medical guideline audit.
Pre-existing Waiting Period	Disease-onset chronology, policy tenure, and moratorium review.
Policy Exclusion	Exact clause retrieval and exception verification.
Financial Limit	Room rent, sub-limit, and itemized billing audit.
Documentation	Document availability comparison and missing-record requisition generation.
Administrative-Procedural	Deadline calculations and justified-delay assessment.
Eligibility-Coverage	Member eligibility, grace period, and portability verification.

Runtime Data Integration

Each investigation uses runtime tables to ground its analysis in policy text, clinical codes, and configured rules.

Runtime File	Purpose
policy_clauses.csv	Policy clauses, waiting periods, exclusions, and exceptions.
denial_reasons.csv	Classification and analysis of insurer denial reasons.
icd10_codes.csv	Diagnosis mapping to ICD-10 codes.
procedure_codes.csv	Medical procedure identification and verification.
nme_exclusions.csv	Review of non-medical expenses against the configured exclusion list.

Overall outcome: The system identifies potentially incorrect claim denials, overlooked policy exceptions, questionable financial deductions, documentation gaps, and procedural issues. Findings should be linked to the relevant policy clause, supporting evidence, and applicable requirements for further verification.

Important note: Numerical limits, statutory interpretations, waiting periods, and legal statements are presented as framework examples and should be verified against current IRDAI regulations, the specific insurance policy, and applicable clinical guidelines before use in real claim decisions.